

Justify your answers and show your work.

(a) Find the inverse on the specified interval express it in the form $y = f^{-1}(x)$. (b) State the range and domain of $f^{-1}(x)$. (c) Graph $f(x)$ and $f^{-1}(x)$. [2 pts each].

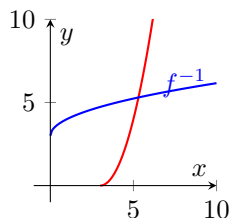
1. $f(x) = x^2 - 6x + 9$ for $x \geq 3$

a.

$$\begin{aligned} y &= (x-3)^2 \\ \Rightarrow \sqrt{y} &= x-3 \\ \Rightarrow f^{-1}(x) &= \sqrt{x} + 3 \end{aligned}$$

b. for Domain: $[0, \infty)$ (f has range $[0, \infty)$), and
Range: $[3, \infty)$ (f has domain $[3, \infty)$).

c.



Find the derivative of the inverse of the following function at the specified point on the graph of the inverse function without finding f^{-1} .

2. $f(x) = 3x^5 - 2x^4 + 1$, at $(1, 0)$ (point on f^{-1}).

$$f'(x) = 15x^4 - 8x \text{ and } f'(0) = 0$$

$(f^{-1})'(0) = \frac{1}{f'(1)} = \frac{1}{0}$ so the derivative does not exist at this point. The slope of the tangency line at this point is vertical.

Use the information about $f(x)$ to answer the following questions. $f(3) = 3$, $f(-4) = \sqrt{5}$, and $f(x)$ has the domain $(-\infty, \infty)$ and range $[\sqrt{5}, \infty)$ [2pts each].

3. Find $f^{-1}(3)$
 $= 3$

4. Find $f^{-1}(\sqrt{5})$
 $= -4$

5. What is the domain of $f^{-1}(x)$
Domain: $[\sqrt{5}, \infty)$.

Perform the indicated operation.

6. $\frac{d}{dx} x^2 e^{-x}$
 $= 2x \cdot e^{-x} + (-e^{-x}) x^2 = (2x - x^2) e^{-x}$
Using the product rule.

7. $\frac{d}{dx} \frac{e^x + 3x}{2}$
 $= \frac{1}{2} (e^x + 3)$

8. $\frac{d}{dx} (2 \ln x + \sqrt{2})$
 $= 2 \frac{1}{x} + 0 = \frac{2}{x}$

$$\begin{aligned} 9. \quad & \frac{d}{dx} \left(\ln(3x^2) - \frac{x}{5} \right) \\ &= 6x \frac{1}{3x^2} - \frac{1}{5} = \frac{2}{x} - \frac{1}{5} \end{aligned}$$

$$\begin{aligned} 10. \quad & \int \left(3e^{-3x} + \frac{1}{x} \right) dx \\ &= -\frac{3}{3} e^{-3x} + \ln|x| + C = -e^{-3x} + \ln|x| + C \text{ (Don't forget the absolute values, } |x| \text{)} \end{aligned}$$

$$\begin{aligned} 11. \quad & \frac{d}{dx} 5^x dx \\ &= \frac{d}{dx} e^{x \ln 5} = \ln 5 \cdot e^{x \ln 5} = 5^x \ln 5 \text{ (or use the formula: } \frac{d}{dx} b^x = b^x \ln b \text{)} \end{aligned}$$

$$\begin{aligned} 12. \quad & \frac{d}{dx} x^{\sqrt{8}} \\ &= \sqrt{8} x^{\sqrt{8}-1} = 2\sqrt{2} x^{2\sqrt{2}-1} \text{ (Don't use } \sqrt{8} \approx 2.8284 \text{)} \end{aligned}$$

$$\begin{aligned} 13. \quad & \frac{d}{dx} x^{\ln x} \text{ (use logarithmic differentiation)} \\ f'(x) &= f(x) \cdot \frac{d}{dx} \ln(f(x)); \frac{d}{dx} \ln(x^{\ln x}) = \frac{d}{dx} \ln^2 x = \frac{1}{x} \ln x + \frac{1}{x} \ln x = \frac{2 \ln x}{x} \\ \Rightarrow f'(x) &= x^{\ln x} \frac{2 \ln x}{x} = x^{\ln x} \ln x \cdot \ln x = x^{\ln x} \ln^2 x \end{aligned}$$

Perform the indicated operation.

$$\begin{aligned} 14. \quad & \int_1^2 2^x dx = \int_1^2 e^{2 \ln x} dx \text{ [6 pts]} \\ &= \frac{1}{\ln 2} e^{2 \ln x} \Big|_1^2 = \frac{1}{\ln 2} (2^x) \Big|_1^2 = \frac{1}{\ln 2} (2^2 - 2) = \frac{2}{\ln 2} \end{aligned}$$

$$\begin{aligned} 15. \quad & \int 3xe^{2x} dx \\ u &= 3x, du = 3 dx; dv = e^{2x} dx, v = \frac{1}{2} e^{2x} \\ &= 3x \cdot \frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} 3 dx = \frac{3xe^{2x}}{2} - \frac{3}{4} e^{2x} + C = \frac{3(2x-1)}{4} e^{2x} + C \end{aligned}$$

$$\begin{aligned} 16. \quad & \int x \ln 2x dx \\ u &= \ln 2x, du = \frac{1}{x} dx; dv = x, v = \frac{x^2}{2} \\ &= \ln 2x \cdot \frac{x^2}{2} - \int \frac{x^2}{2} \frac{1}{x} dx = \ln 2x \cdot \frac{x^2}{2} - \frac{1}{4} x^2 + C \end{aligned}$$

$$\begin{aligned} 17. \quad & \int x^2 \cos 4x dx \\ u &= x^2, du = 2x dx; dv = \cos 4x, v = \frac{1}{4} \sin 4x \\ &= \frac{x^2 \sin 4x}{4} - \frac{2}{4} \int x \sin 4x dx. \text{ Using integration by parts again, we get } \int x \sin 4x dx = \frac{\sin 4x}{16} - \frac{x \cos 4x}{4} \\ \Rightarrow &= \frac{x^2 \sin 4x}{4} - \frac{\sin 4x}{32} - \frac{x \cos 4x}{8} \end{aligned}$$